

A Five-Year Study of Slotted Eye Plaque Radiation Therapy for Choroidal Melanoma: Near, Touching, or Surrounding the Optic Nerve

Paul T. Finger, MD,^{1,2,3} Kimberly J. Chin, OD,¹ Lawrence B. Tena, MD^{1,2,3}

Objective: To evaluate slotted eye plaque radiation therapy for choroidal melanomas near the optic disc.

Design: A clinical case series.

Participants: Twenty-four consecutive patients with uveal melanomas that were near, touching, or surrounding the optic disc.

Intervention: Slotted eye plaque radiation therapy.

Main Outcome Measures: Recorded characteristics were related to patient, clinical, and ophthalmic imaging. Data included change in visual acuity, tumor size, recurrence, eye retention, and metastasis.

Results: From 2005 to 2010, 24 consecutive patients were treated with custom-sized plaques with 8-mm-wide, variable-depth slots. Radiation doses ranged from 69.3 to 163.8 Gy (mean, 85.0 Gy) based on delivering a minimum tumor dose of 85 Gy. All treatments were continuously delivered over 5 to 7 days. Mean patient age at presentation was 57 years. Tumors were within 1.5 mm of the optic nerve (n = 3, 13%), juxtapapillary (n = 6, 25%), touching ≥ 180 degrees (n = 7, 29%), or circumpapillary (n = 8, 33%). Ultrasound revealed dome-shaped tumors in 79% of patients, collar-button tumors in 17% of patients, irregular tumor in 1 patient (4%), and intraneural invasion in 2 patients. Mean initial largest basal dimension was 11.0 mm (standard deviation [SD] $\pm$ 3.5 mm; median, 11.4 mm; range, 5.9–16.4 mm). Mean initial tumor thickness was 3.5 mm (SD $\pm$ 1.7 mm; median, 3.0 mm; range, 1.4–6.9 mm). Initial visual acuities were a median 20/25 (range, 20/20 to hand motions) and decreased to a median 20/40 (range, 20/20 to no light perception). At a mean follow-up of 23 months, 12 patients required periodic intravitreal bevacizumab to suppress radiation optic neuropathy (RON) or maculopathy. To date, there has been a 100% local control rate. No patients have required secondary enucleation for recurrence or neovascular glaucoma. No patients have developed metastasis.

Conclusions: Slotted plaque radiation therapy allows peripapillary, juxtapapillary, and circumpapillary choroidal melanomas (and a safety margin) to be included in the radiation targeted zone. Normalization of the plaque position beneath the tumor appears to increase RON and improve local control.

Financial Disclosure(s): The author(s) have no proprietary or commercial interest in any materials discussed in this article. *Ophthalmology* 2012;119:415–422 © 2012 by the American Academy of Ophthalmology.



Optic nerve anatomy presents a unique obstacle for radioactive plaque placement. The optic disc face (papilla) has a mean diameter of 1.8 mm and is surrounded by choroid. Thus, choroidal melanomas can touch or encircle the disc. Also, as the optic nerve exits the eye, it is enveloped by the optic nerve sheath, thus expanding its total width to a 5- to 6-mm optic nerve sheath diameter (ONSD).¹ This difference creates an offset to proper plaque placement (Fig 1A).^{2–4}

For example, when a standard ophthalmic plaque is placed against the retrobulbar optic nerve sheath, its most posterior edge will only reach to 1.5 mm from the edge of the optic disc and the juxtapapillary choroidal melanoma (Fig 1B). Further, any melanoma that extends within 3.5 mm of the optic disc cannot be surrounded by the standard 2.0-mm radiation margins used for extrapapillary tumors.

However, despite this anatomic handicap, plaque radiation therapy for selected juxtapapillary melanomas historically offers up to 75% local control.^{5,6} Success has been attributed to plaque-notching, radiation side-scatter, posterior plaque tilting, and adjuvant argon laser or transpupillary thermotherapy (TTT).^{4,5,7–15}

Because of relatively poor local control with the aforementioned modalities, many eye cancer specialists currently recommend enucleation or specialized forms of external beam radiation therapy (EBRT) (e.g., proton, cyberknife, and stereotactic radiosurgery) for select juxtapapillary and most circumpapillary choroidal melanomas.^{16–20}

In 2005, the first 8-mm-wide and 8-mm-deep slot was cut from the posterior aspect of an 18-mm Collaborative Ocular Melanoma Study (COMS)-type gold plaque shell.²¹

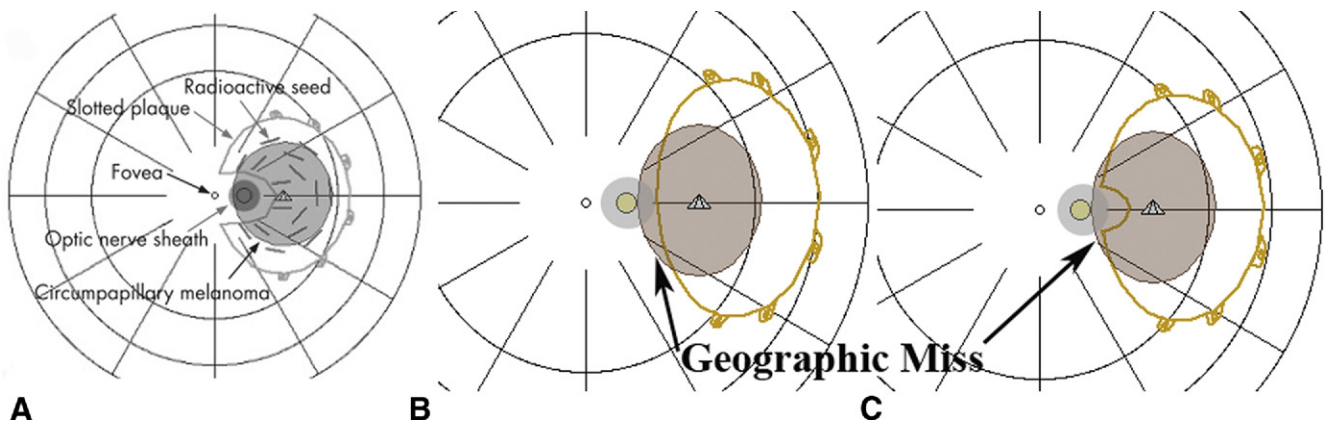


Figure 1. A, Anatomic relationships are described. B, Graphic image of a hypothetical juxtapapillary tumor partially covered by a round Collaborative Ocular Melanoma Study type plaque placed against the optic nerve sheath. Geographic miss is indicated by the arrow. C, In that the 4-mm-wide, 2-mm-deep “standard” notch cannot fit around the optic nerve, in addition to the margin of tumor along the posterior edge, there is an additional gap of geographic miss created by the notch (arrow).

Termed *Finger’s slotted eye plaque*, the shell was created to allow the retrobulbar optic nerve (ONSD) to enter the plaque, allow the plaque to bypass that obstruction, and extend beyond the opposite side of the optic disc.²¹ With a slotted plaque in place, the entire tumor and 2-mm tumor-free margin were located beneath the plaque. Modulating the strength and position of radionuclide seeds (within the altered plaque) allowed us to “fill in” the radiation dose within the slot. Encompassing the entire tumor and free margin allows for normalized treatment of this subset of choroidal melanomas.

This study examines our experience with slotted plaque brachytherapy for choroidal melanomas that are near, touching, or surrounding the optic disc. With more than 5 years of experience, this study examines how the Finger’s slotted plaque technique affects local tumor control, radiation side effects, and visual acuity.

Patients and Methods

This study conformed to the Declaration of Helsinki and the US Health Insurance Portability and Privacy Act of 1996. It was approved by the internal review board and ethics committees of The New York Eye Cancer Center.

Patient Selection and Informed Consent

Patients selected for this series had choroidal melanomas within 1.5 mm of the edge of the optic disc. This included peripapillary, juxtapapillary, and circumpapillary choroidal melanomas. Informed consent focused on the choice of potential for vision, local tumor control, and preservation of life. Patients were specifically informed that posterior migration of the plaque’s targeted zone (around the optic disc) increased the dose to the nerve and the risk for radiation optic neuropathy (RON)-related vision loss.^{22–27} They were also told that slotted plaque therapy was the only method of plaque placement that would include the entire tumor and safety margin beneath the plaque. We noted that failure to use a safety margin has been associated with failure of local con-

trol.^{6,28} Patients were also informed that all known forms of EBRT techniques also included the optic nerve within the irradiated zone and risked RON.²³ Patients were told that, when compared with enucleation, all the aforementioned radiation therapy techniques offered some potential for vision retention. Lastly, each patient was informed about the number of patients who had been treated with slotted radioactive plaque therapy and their outcomes.

Patient characteristics recorded were age, sex, eye (right or left), medical history (hypertension or diabetes mellitus), best-corrected visual acuity (using early diabetic retinopathy treatment charts), pupillary findings, and ophthalmic examination findings (anterior segment findings, intraocular pressure, lens status). Tumor characteristics included shape, size (height and basal dimensions), presence of perineural invasion, distance to optic nerve, and foveal avascular zone. Radiation characteristics recorded were plaque size, slot size, radioactive isotope, tumor dose, length (hours) of radiation exposure, dose rate, and dose to other structures (lens, fovea, optic nerve, opposite retina, and inner sclera).

Patients were followed with ophthalmic examination, comparative funduscopy, ultrasound, and ophthalmic imaging (photography, optical coherence tomography) every 3 to 4 months after brachytherapy. Fluorescein angiography was typically performed every 6 months. Tumor height (by ultrasonography) and basal dimensions (by funduscopy and photography) were compared at each visit.

Local Recurrence

The determination of local recurrence can be complicated by acute post-brachytherapy tumor edema, resolution of peritumoral retinal detachment, and stimulation of the adjacent retinal pigment epithelium. For the purpose of this study, local recurrence was defined as >0.5 mm of apical tumor growth (by ultrasonography) or >1 mm of vascularized marginal growth (by comparative funduscopy, photography, and retinal angiography).

Systemic metastasis was evaluated by clinical examination supplemented by positron emission tomography/computed tomography, computed tomography, or magnetic resonance imaging. Patients underwent whole-body positron emission tomography/computed tomography as initial staging, with restaging every 6 months for the first 3 years and every year thereafter.

Plaque Definitions

Standard Plaques. In 1985, the COMS used Trachsel Dental Studios (Rochester, MN) to produce gold eye plaques with diameters ranging from 12 to 20 mm and graduated edges (lips) and silicone inserts designed to offset radioactive seeds from the sclera and the gold. The lips or sidewalls of the plaque functioned as a partial shield to the sides of the plaque. The gold backing of the plaque blocks the posterior dose, creating a unidirectional source into the eye.

Notched Plaque. Trachsel Studios and EyePhysics LLC (Los Alamitos, CA) produce COMS-style plaques that have a 4-mm-wide and 2-mm-deep walled notch at their posterior margin. Notches are made to accommodate the optic disc but are too small to allow entry of the orbital portion of the optic nerve (Fig 1C). Therefore, plaques with standard-sized notches will not cover juxtapapillary tumors adequately.

Slotted Plaque. To date, slotted plaques have been created from standard COMS-type plaques. The slot width is 8-mm wide to accommodate the ONSD and adexa. The slot depth is calculated (Table 1, available at <http://aaojournal.org>) and cut out by a local dental laboratory.

Radiation Definitions

Targeted Zone. In treatment of uveal melanoma, the targeted zone typically includes all visible tumor plus a 2- to 3-mm margin of safety.

Margin of Safety. Radiation treatment involves calculating dose to the tumor plus a surround of normal-appearing tissue called the “margin of safety” or “treatment margin.” Treatment margins are added to not miss the tumor and possible microscopic tumor extension. In addition, patients’ eyes are known to move during irradiation, causing submillimeter plaque movements.²⁹ It is reasonable to assume that without a margin of safety, part of the cancer would occasionally move out of the targeted zone, receive less irradiation than planned, and survive treatment.

Geographic Miss. Geographic miss occurs when there is a misalignment between the tissue volume treated (e.g., area beneath the plaque) and the tumor plus free margin (Fig 1B, C).

Tumor Definitions

Extrapapillary choroidal melanoma is a tumor that extends no closer than 3.5 mm from the optic disc margin. With perfect plaque placement, the edge of the round COMS-type plaque touches the optic nerve sheath and provides a standard 2-mm posterior margin of safety. Thus, slotting of the plaque is not needed, and there is little risk for geographic miss.

Peripapillary choroidal melanoma is a tumor that extends between 1.5 and 3.5 mm of the optic disc. In this case, even if a standard round COMS-type plaque were perfectly placed to touch the optic nerve sheath, there would be less than a 2-mm posterior treatment margin of safety. An 8-mm-wide, 1- to 2-mm-deep slot will allow for a portion of the optic nerve sheath to enter the plaque and extend the posterior margin of safety.

Juxtapapillary choroidal melanoma is a tumor that extends to touch the optic disc margin. There is no way that a round or notched plaque can cover the posterior extent of these cancers (Fig 1B and C). With a rounded plaque placed perfectly against the optic nerve sheath, its posterior edge is ≥ 1.5 mm from the posterior edge of the tumor. In this case, geographic miss of the main tumor body is guaranteed. In these cases, slot depth is calculated to accommodate as much optic nerve sheath as needed to extend the plaque beneath the entire tumor and a 2-mm free margin.

Circumpapillary choroidal melanoma is a tumor that completely surrounds the optic nerve. Encircling the nerve, the tumor may extend anteriorly for variable distances and can grow to cover or invade the optic nerve disc. When calculated to a correct depth, the slot can allow the entire tumor and free margin to be covered by the ophthalmic plaque. In general, these cases may require rather deep slot plaques to include the entire tumor and free margin within the targeted zone (plus 10 mm to allow for access to the suture eyelets). Patients with intraneural extension greater than half of the depth of a COMS plaque may not be amenable to slotted plaque therapy.

Slotted Plaque Construction

Slot Width. Slotted plaques were designed to compensate for the epibulbar optic nerve obstruction or ONSD. The slot accommodates for and extends around the optic nerve sheath. Slotted plaque construction has been described. In summary, an 8-mm-wide (transverse) slot was cut from standard COMS plaques centered at their most posterior margin. The 8-mm dimension was determined from 3-dimensional ultrasound imaging and review of the radiographic literature.³⁰ Although we initially measured each optic nerve, we have subsequently found that an 8-mm slot width has been ample to accommodate all ONSDs treated in this series.

Slot Depth. The second critical dimension is slot depth (longitudinal diameter). The depth of the slot was determined for each tumor to produce a 2- to 3-mm margin of safety beyond the tumor’s posterior edge. Thus, the optic nerve (within its sheath) was able to enter the plaque and synchronously extend the posterior plaque edge beyond the visible tumor to include a standard 2-mm margin of safety. Figure 2 demonstrates how variable-depth slots can be used to extend the posterior margin of the plaque beyond a juxtapapillary, 180-degree, and complete circumpapillary tumor. Table 1 (available at <http://aaojournal.org>) provides a reference table for these 3 types of tumors and shows slot depths and plaque sizes that are required.

Specific Slot Depth Calculations

Given. The peripapillary edge of the 5-mm orbital optic nerve sheath can be calculated as follows: One half the ONSD or 2.5 mm minus 0.9 mm or $\frac{1}{2}$ a disc diameter = 1.6 mm. Thus, 1.6 plus 1.8 (optic disc diameter) plus 1.6 mm equals a 5-mm ONSD.

Extrapapillary Tumor. When the tumor is ≥ 3.5 mm from the disc margin, it does not require a slot. Both the tumor and a 2-mm free margin can be placed within the targeted zone using a standard round plaque.

Peripapillary Tumor. When the tumor is < 3.5 mm from the optic disc, the slot depth should be equivalent to the amount of posterior migration needed to include the 2-mm free margin. For example, if the tumor extends to 1 mm from the optic disc margin, a perfectly placed round plaque would not even cover 0.6 mm because of the optic nerve sheath obstruction. Therefore, the calculation would include 0.6 mm plus a 2-mm free margin = 2.6-mm slot depth.

Juxtapapillary Tumor. When the tumor is merely touching the optic disc, the slot depth should be calculated as 1.6 mm to accommodate for the ONSD, plus 2 mm for the safety margin, for a total of 3.6, or 4.6 mm for a 3-mm margin.

Circumpapillary Tumor. If a hypothetical 12×12-mm tumor is circumpapillary and extends 3 mm beyond the posterior (opposite) margin of the optic disc, then the entire ONSD of 5 mm, plus 3 mm of tumor, plus 2 mm of safety margin is accommodated within a 10-mm slot depth. Then add 10 mm anterior to the slot (for ease of plaque insertion) for a minimum 20-mm plaque size (Fig 2C).

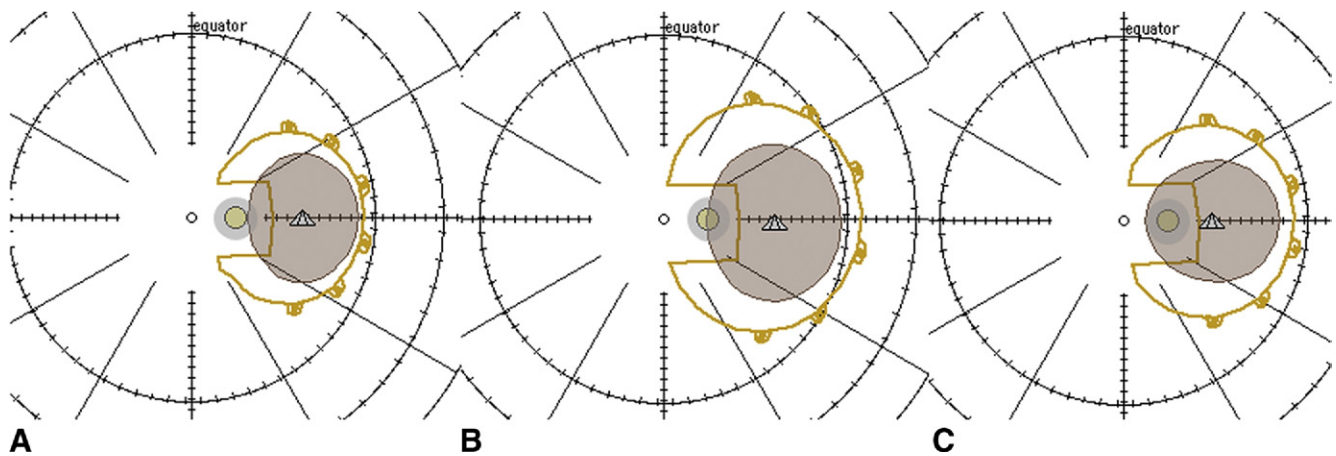


Figure 2. Retinal drawings show how variable-depth slots can be used to extend the posterior margin of the plaque beyond juxtapapillary (A), 180-degree (B), and complete circumpapillary (C) tumors.

Plaque Diameter. Commercially available plaque diameters range from 10 to 22 mm. For ease of plaque insertion (access to the plaque eyelets), a minimum of 10 mm of plaque diameter is needed from the apex of the slot, along the longitudinal axis toward the anterior plaque margin. Thus, for a 4-mm slot depth, the minimum plaque diameter should be 14 mm. For example, we have provided a quick chart of slot and plaque sizes (Table 1, available at <http://aaojournal.org>).

Seed Orientation and Radiation Dose. To maximize the tumor dose and minimize irradiation of normal ocular structures, radioactive seeds were manually affixed within the plaque (Appendix 1, available at <http://aaojournal.org>). The position or distribution of radioactive seeds was adjusted to conform to tumor size and location. Specifically, they were positioned to ensure adequate dose coverage of the tumor in the gap created by the slot. Seed locations were photodocumented, digitized, and entered into the treatment program system. Thus, the dose calculations for

tumor coverage were calculated by certified medical physicists to provide a minimum tumor tissue dose using a brachytherapy treatment planning computer program to account for custom placement of seeds.

Slotted Plaque Insertion

Slotted plaques were inserted in a manner similar to standard COMS plaques for extrapapillary choroidal melanoma. However, temporary disinsertion of both rectus and oblique muscles were more commonly required because of the posterior location of the tumors. Because slotted plaques straddle the retrobulbar optic nerve, this structure acted as a reference point for proper plaque placement. Intraoperative ultrasound imaging was used to confirm plaque placement (Fig 3).^{3,4,29,31–33} No acute optic nerve infarction or retrobulbar hemorrhage was noted in this series.

Results

Slotted plaques were constructed and used to deliver brachytherapy for the treatment of 24 patients with choroidal melanoma. The mean patient age at presentation was 57 years (median, 61 years; range, 22–83 years). There were 12 male and 12 female patients, and 12 right eyes were affected. One patient had diabetes mellitus, and 11 patients had hypertension at the time of melanoma treatment. There were 12-T1, 11-T2, 1-T3, and 0-T4 tumors according to the 7th edition of the American Joint Committee on Cancer–International Union Against Cancer classification system for uveal melanoma (Table 2, available at <http://aaojournal.org>).³⁴

Choroidal melanomas were peripapillary (within 1.5 mm of the optic nerve) in 3 patients (13%), juxtapapillary (touching up to 180 degrees of the optic disc) in 6 patients (25%), touching ≥ 180 degrees in 7 patients (29%), or circumpapillary (surrounding 360 degrees of the optic disc) in 8 patients (33%).

All choroidal melanomas in this series were treated with plaques containing an 8-mm-wide slot and variable-depth slots. In this series, the mean plaque size was 17 mm (range, 12–20 mm), with a mean slot depth of 5.7 mm (range, 3–8 mm). Tumor apex prescription doses ranged from 69.3 to 163.8 Gy (mean, 85.0 Gy) based on calculations delivering a minimum of 85 Gy tumor dose to the tumor within the slot. This resulted in a mean scleral dose of 262.2 Gy (range, 165.1–509.4 Gy). Although this is a relatively high value, none of the eyes were noted to have scleral thinning on

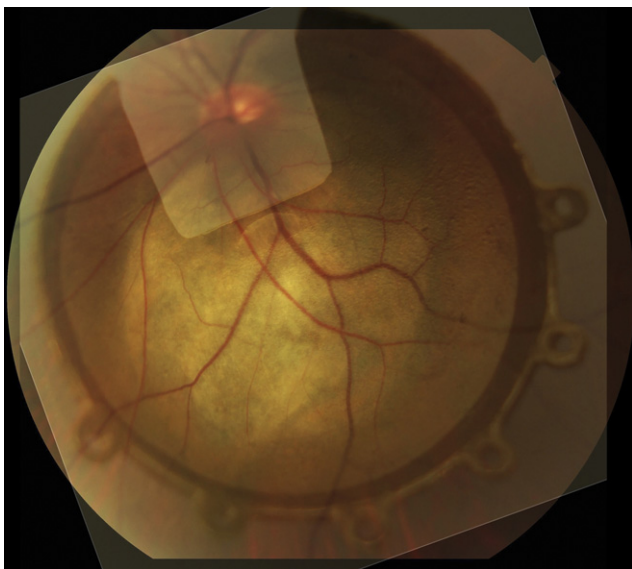


Figure 3. Fundus photograph of a circumpapillary choroidal melanoma. A slotted plaque is superimposed on the photograph to depict the outline of where the plaque would hypothetically rest around the tumor and optic nerve sheath.

Table 3. Treatment Outcomes for Slotted Plaque Brachytherapy

Treatment Outcomes	Baseline No. (%)	Last Findings No. (%)
Visual acuity (n = 22)*		
Median acuity	20/25	20/40
>20/200	20 (20/22, 91%)	17 (17/22, 77%)
≥20/40	16 (16/22, 73%)	12 (12/22, 55%)
Loss of ≥2 lines	—	9 (9/22, 41%)
Complications		
Neovascular glaucoma (n = 24)	—	0
Radiation cataract (n = 18 – phakic)		1 (1/18, 5.5%)
Radiation Optic Neuropathy (RON)		15 (15/24, 62.5%)
Radiation maculopathy (RM)		9 (9/24, 37.5%)
Treatment with anti-VEGF (n = 22)†		12 (12/22, 54.5%)
Enucleation	—	0
Tumor recurrence	—	0
Metastasis/death	—	0
Follow-up (mos)		
Median; mean ± SD (range)		22; 23 ± 14.7 (4–64)

RON = radiation optic neuropathy; VEGF = vascular endothelial growth factor.

*Limited to those patients with at least 6 months of follow-up data available.

†Excluding 2 patients treated with anti-VEGF for wet macular degeneration (1) and preoperative subretinal fluid (1).

ultrasonography, to have extrascleral extension, or to require a scleral patch graft. All treatments were continuously delivered over 5 to 7 days. After treatment, patients were followed every 3 to 4 months for a mean of 23 months (standard deviation [SD] ± 14.7 months; median, 22 months; range, 4–64 months) from the time of plaque brachytherapy.

Ultrasound imaging revealed dome (n = 19, 79%), collar-button (n = 4, 17%), and irregular (n = 1, 4%) tumors. Two patients (8.7%) had intraneural invasion. The mean initial largest basal dimension was 11.0 mm (SD ± 3.5 mm; median, 11.4 mm; range, 5.9–16.4 mm). The mean initial tumor thickness was 3.5 mm (SD ± 1.7 mm; median, 3.0 mm; range, 1.4–6.9 mm). Follow-up ultrasound imaging revealed mean tumor regression to 2.4 mm (SD ± 1.5 mm; median, 1.7 mm; range, 0.6–5.0 mm). Ophthalmic ultrasound, magnetic resonance imaging, or computed tomography was used to evaluate affected optic nerves that were obscured by tumor.

Visual Acuity

Reporting visual acuity is limited to the 22 patients who had at least 6 months of follow-up at the time of data collection (Table 3). Initial visual acuities were a median of 20/25 (range, 20/20 to no light perception) and decreased to a median of 20/40 (range, 20/20 to no light perception) with a mean follow-up of 25 months (range, 6–64 months). Twenty patients (20/22, 91%) had a baseline acuity >20/200, and 17 patients (17/22, 74%) retained >20/200 visual acuity at the last follow-up examinations. Sixteen patients (73%) had ≥20/40 vision at baseline, which decreased to 12 patients (55%) at the last visit. Nine of 22 patients followed for at least 6 months lost >2 lines of vision. Loss of visual acuity in these patients was attrib-

uted to anterior RON (7/9, 78%) and radiation maculopathy (1/9, 11%). One patient (11%) progressed to no light perception because of posterior ischemic optic neuropathy.

An analysis of visual acuity over time was performed. We divided patients into 2 subgroups viewed at 3 time intervals (12, 24, and 36 months). We examined those patients who were able to maintain ≥20/200 and those who started with ≥20/40 and were able to maintain that level of visual acuity. Visual acuity at 12, 24, and 36 months (±2 months) were as follows. At 12 months (n = 19), of the 18 patients who started with ≥20/200 vision, 17 were able to maintain ≥20/200. Of the 14 patients who started with ≥20/40 vision, 11 were able to maintain ≥20/40. Three patients started with <20/40 vision and improved to ≥20/40 at 12 months. At 24 months (n = 12), of the 11 patients who initially had ≥20/200 vision, 8 had ≥20/200 at this time point. Of the 8 (of 12) patients who initially had ≥20/40 vision, 3 were able to maintain ≥20/40. At 36 months (n = 5), of the 5 patients who initially had ≥20/200 vision, 4 were able to maintain ≥20/200. Of the 4 patients who started with ≥20/40 vision, 2 were able to maintain ≥20/40.

Likewise, these 22 patients were examined for the presence of radiation maculopathy and RON and were staged according to the Ophthalmic Radiation Oculopathy Vision Prognosis Classification (Table 4, available at <http://aaojournal.org>). Sixteen of 22 patients (73%) developed stage 4 retinopathy primarily characterized by vitreous hemorrhage and RON. No patients had stages 1 to 3 retinopathy, and 5 patients had stage 0 (no retinopathy) at the time of last follow-up. One additional patient with “stage 0” developed wet macular degeneration.

Radiation optic neuropathy was the most common cause of treatment-related vision loss. Radiation optic neuropathy developed in 15 of 24 patients (62.5%), and radiation maculopathy developed in 9 of 24 patients (37.5%). Fourteen patients have been treated with intravitreal anti-vascular endothelial growth factor (VEGF) therapy to suppress RON in 12 of 22 patients (54.5%). Anti-VEGF therapy was typically initiated at the first clinical sign of neuropathy or subjective worsening of vision (not prophylactic). In addition, intravitreal ranibizumab was administered for synchronous wet macular degeneration in 1 patient and for a pre-brachytherapy subfoveal exudative retinal detachment in 1 patient.

Of the 18 patients who were phakic at the time of treatment, 1 (5.5%) developed a radiation cataract and 1 had preoperative cataract. Twenty patients were assessed for pupillary function. At last follow-up, 50% of patients (n = 10/20) were noted to have a relative afferent pupillary defect in their affected eye, developing at a mean of 9 months after plaque brachytherapy (range, 3–21 months) in 9 patients. One patient had a preoperative relative afferent pupillary defect.

Local Control and Metastasis

The primary goal of local treatment, control, was achieved in all 24 patients. Local control was defined as the lack of tumor growth, apical (by ultrasonography) or marginal (by comparative funduscopy and photography). B-scan revealed a mean initial tumor thickness of 3.5 mm (SD ± 1.7 mm; median, 3.0 mm; range, 1.4–6.9 mm), and at last follow-up tumors regressed to a mean 2.4 mm (SD ± 1.5; median, 1.7 mm; range, 0.6–5.0 mm). No patient was noted to have apical or marginal tumor recurrence. One patient underwent panretinal photocoagulation for retinal ischemia and neovascularization. It is important to note that adjuvant tumor TTT was not needed to obtain local control for any patient in this series. In addition, no patient developed iris neovascularization (neovascular glaucoma) or required secondary enucleation. No patient developed metastasis during the study interval as monitored by periodic abdominal radiographic imaging.

Table 5. Published Studies of Treatment for Juxtapapillary Choroidal Melanoma

Author	Radiation	Dose	Adjuvant Laser	Patients	Follow-up	Local Control	Metastasis	Enucleation
Somani et al ³⁶	Stereotactic radiosurgery	70 Gy	None	64	26 mos	94%	12%	11%
Roberge et al ³⁷	External beam cobalt-60	65 Gy	None	26	60 mos	84%	26%	19%
De Potter et al ¹²	Multiple plaque sources	85 Gy	Some TTT	93	78 mos	85%	12%	22%
Sagoo et al ¹⁴	Multiple plaque sources	85 Gy	Some TTT	28	46 mos	86%	4%	25%
Sagoo et al ¹⁵	Multiple plaque sources	85 Gy	Some TTT	141	56 mos	90%	13%	23%
Sagoo et al ⁶	Multiple plaque sources	80 Gy	Some TTT	650	60 mos	86%	11%	n/a
Lommatzsch and Lommatzsch ⁵	Ruthenium-106	100 Gy	Some xenon	39	48 mos	90%	13%	8%
Lommatzsch et al ²²	Ruthenium-106	100 Gy	n/a	93	41 mos	85%	13%	n/a
Present study	Palladium-103	85 Gy	None	24	22 mos	100%	0%	0%

Transpupillary thermotherapy (TTT) included infrared, argon, and krypton in some cases. Multiple sources included cobalt-60, iodine-125, ruthenium-106, and iridium-192.

Discussion

Slotted plaque radiation therapy can be used to control peripapillary, juxtapapillary, and circumpapillary choroidal melanomas. Incorporating the optic nerve (within the plaque) allows for normalization of plaque placement to include the entire tumor and safety margin within the targeted zone. The optic nerve also serves to localize the posterior aspect of the plaque. As a result, slotted plaque irradiation yielded local control rates better than that expected with standard plaque radiation of extrapapillary choroidal melanomas. Further, in consideration of this study's 5-year results, it is our impression that the use of slotted plaques will improve local control compared with notched plaque designs, tilted plaques, or treatment with adjuvant TTT.

It is important to note that in the past, plaque radiotherapy techniques (for peripapillary melanomas) relied on laser photocoagulation (infrared-TTT, argon, xenon-arc) to treat portions of the tumor known to be outside the irradiated zone (Table 5).^{5,6,9,11,13-15} In contrast, slotted plaque radiation therapy encompasses the entire tumor plus free margin and does not require an adjuvant laser.

Notched versus Slotted Eye Plaques

By definition, notches have never been able to accommodate the optic nerve sheath within the plaque. In fact, the typical 4-mm-wide, walled notch creates a small gap between the posterior plaque margin and the optic nerve sheath (Fig 1C). When present, the side wall within the notch further shields the tumor from irradiation. Successful notched plaque therapy depends on tilt and side-scatter irradiation.

In contrast, slotted plaques are wide enough to allow the retrobulbar optic nerve into the plaque (to a measured depth). The posterior plaque margin extends beyond the tumor's posterior margin. Furthermore, it has no posterior side wall (within the slot) to block radiation to the adjacent tumor. In contrast with notched plaques, slotted plaques do not rely on side-scatter irradiation. They allow the entire tumor to be surrounded by radioactive seeds within the target zone (Fig 1A).

Radioactive Seed versus Solid Sources

Slotted plaques are best suited for low-energy ¹⁰³Pd or ¹²⁵I sources because radioactive seed strength and location can be modulated to conform to the location and size of the tumor.³⁵ Furthermore, they add flexibility needed to fill in volume beneath the slot. In contrast, ruthenium-106 plaques are not designed to irradiate within the minimum 8-mm-wide variable-depth slot. Although such a configuration might be possible, cutting into a ¹⁰⁶Ru plaque would create a serious radiation safety hazard. Lastly, the penumbra for ¹⁰⁶Ru is so sharp, it is not able to fill in the volume within the slot.

Slotted Plaques versus External Beam Radiation Therapy

Compared with stereotactic radiosurgery, the cyberknife, and proton beam, slotted plaque therapy offers the most conformal (less irradiation to normal anterior ocular, adnexal, and orbital structures) tumor treatment (Table 5). For example, Somani et al³⁶ published a study of stereotactic radiotherapy for juxtapapillary melanomas performed at Princess Margaret Hospital in Toronto, Canada. They reported on 64 patients with tumors within 2 mm of the optic disc treated to 70 Gy in 5 fractions over 10 days. With a median follow-up of 26 months, there were 6% local recurrences, 28% neovascular glaucomas, 45% cataract, 80% retinopathy, and 52% optic neuropathy.³⁶ In comparison, this series of slotted plaque brachytherapy was associated with 0% local recurrence, 0% neovascular glaucoma, 5.5% radiation cataract, 39% radiation maculopathy, and 62.5% RON (with a mean follow-up of 2 years).

This comparative distribution of side effects can be related to relative dose to the tumor and normal ocular structures. For example, both slotted plaque brachytherapy and EBRT techniques deliver similar amounts of radiation to the tumor and anterior optic nerve. Therefore, both techniques demonstrate high rates of local control and optic neuropathy. However, unlike external beam techniques, slotted plaque therapy does not require an anterior ocular entry dose. Therefore, slotted plaque radiation therapy yielded less or no neovascular glaucoma, radiation cataract, eyelash loss, and dry eye.³⁷ It is also important to consider that the

orbital optic nerve can be shielded by the gold of the plaque, suggesting caution in treatment of those occasional cases in which intraneural invasion is found.^{17,38}

Radiation Optic Neuropathy

High rates of RON may be acceptable in that slotted plaque radiation therapy is typically used as an eye- and vision-sparing alternative to enucleation. In this study, patients only lost a median of 2 lines of visual acuity, decreasing from a median of 20/25 to 20/40 during the duration of this study. In consideration of the high rates of RON, our findings of vision preservation have been attributed to the discovery that periodic administration of intravitreal anti-VEGF medications can suppress radiation vasculopathy and optic neuropathy.^{24,39–42}

Local Control

Most important, no tumor primarily treated with slotted plaque brachytherapy has recurred with a median follow-up of 22 months (range, 6–64 months). In contrast, Somani et al³⁶ reported an actuarial 6% regrowth rate at 26 months, and Roberge et al³⁷ reported a 16% regrowth rate at 60 months. The results of Lommatzsch and Lommatzsch,⁵ De Potter et al,¹² Sagoo et al,^{6,14,15} and Lommatzsch et al²² are not comparable because of synchronous adjuvant laser therapy. Although they cumulately noted a 13% failure rate at a mean of 55 months, these reports are cumulatively from 6 articles on 2 data sets (Table 5).

In conclusion, in this series, we attribute our favorable local control results to the elimination of geographic miss (as occurs with round or notched plaques) and normalization of the targeted zone to include the entire tumor and free margin. Therefore, this study suggests that Finger's slotted plaque design improves our ability to match tumor and plaque location by overcoming the obstacle of the orbital portion of the optic nerve. Further, our results suggest that normalization of the plaque location to include the tumor and a free margin increases the dose to both the optic nerve and the choroidal melanoma. More long-term follow-up will be required to compare this technique with plaque plus laser and EBRT techniques.

References

- Garcia JP Jr, Garcia PT, Rosen RB, Finger PT. A 3-dimensional ultrasound C-scan imaging technique for optic nerve measurements. *Ophthalmology* 2004;111:1238–43.
- Finger PT. Radiation therapy for choroidal melanoma. *Surv Ophthalmol* 1997;42:215–32.
- Harbour JW, Murray TG, Byrne SF, et al. Intraoperative echographic localization of iodine 125 episcleral radioactive plaques for posterior uveal melanoma. *Retina* 1996;16:129–34.
- Almony A, Breit S, Zhao H, et al. Tilting of radioactive plaques after initial accurate placement for treatment of uveal melanoma. *Arch Ophthalmol* 2008;126:65–70.
- Lommatzsch PK, Lommatzsch R. Treatment of juxtapapillary melanomas. *Br J Ophthalmol* 1991;75:715–7.
- Sagoo MS, Shields CL, Mashayekhi A, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma: tumor control in 650 consecutive cases. *Ophthalmology* 2011;118:402–7.
- Harbour JW, Meredith TA, Thompson PA, Gordon ME. Transpupillary thermotherapy versus plaque radiotherapy for suspected choroidal melanomas. *Ophthalmology* 2003;110:2207–15.
- Lean EK, Cohen DM, Liggett PE, et al. Episcleral radioactive plaque therapy: initial clinical experience with 56 patients. *Am J Clin Oncol* 1990;13:185–90.
- Brovkina AF, Zarubei GD, Fishkin IuG. Validation of the use of brachytherapy in uveal melanomas of juxtapapillary localization [in Russian]. *Vestn Oftalmol* 1991;107:41–4.
- Packer S, Stoller S, Lesser ML, et al. Long-term results of iodine 125 irradiation of uveal melanoma. *Ophthalmology* 1992;99:767–74.
- Oosterhuis JA, Jorner-de Korver HG, Kakebeeke-Kemme HM, Bleeker JC. Transpupillary thermotherapy in choroidal melanomas. *Arch Ophthalmol* 1995;113:315–21.
- De Potter P, Shields CL, Shields JA, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma: visual acuity and survival outcome. *Arch Ophthalmol* 1996;114:1357–65.
- Seregard S, Landau I. Transpupillary thermotherapy as an adjunct to ruthenium plaque radiotherapy for choroidal melanoma. *Acta Ophthalmol Scand* 2001;79:19–22.
- Sagoo MS, Shields CL, Mashayekhi A, et al. Plaque radiotherapy for choroidal melanoma encircling the optic disc (circumpapillary choroidal melanoma). *Arch Ophthalmol* 2007;125:1202–9.
- Sagoo MS, Shields CL, Mashayekhi A, et al. Plaque radiotherapy for juxtapapillary choroidal melanoma overhanging the optic disc in 141 consecutive patients. *Arch Ophthalmol* 2008;126:1515–22.
- Scuderi G. Juxtapapillary malignant melanoma of the choroid involving the optic disk. *Ophthalmologica* 1973;166:349–52.
- Shammas HF, Blodi FC. Peripapillary choroidal melanomas: extension along the optic nerve and its sheaths. *Arch Ophthalmol* 1978;96:440–5.
- Emara K, Weisbrod DJ, Sahgal A, et al. Stereotactic radiotherapy in the treatment of juxtapapillary choroidal melanoma: preliminary results. *Int J Radiat Oncol Biol Phys* 2004;59:94–100.
- Zorlu F, Selek U, Kiratli H. Initial results of fractionated CyberKnife radiosurgery for uveal melanoma. *J Neurooncol* 2009;94:111–7.
- Fernandes BF, Weisbrod D, Yucel YH, et al. Neovascular glaucoma after stereotactic radiotherapy for juxtapapillary choroidal melanoma: histopathologic and dosimetric findings. *Int J Radiat Oncol Biol Phys* 2011;80:377–84.
- Finger PT. Finger's "slotted" eye plaque for radiation therapy: treatment of juxtapapillary and circumpapillary intraocular tumours. *Br J Ophthalmol* 2007;91:891–4.
- Lommatzsch PK, Alberti W, Lommatzsch R, Rohrwacher F. Radiation effects on the optic nerve observed after brachytherapy of choroidal melanomas with 106Ru/106Rh plaques. *Graefes Arch Clin Exp Ophthalmol* 1994;232:482–7.
- Gragoudas ES, Li W, Lane AM, et al. Risk factors for radiation maculopathy and papillopathy after intraocular irradiation. *Ophthalmology* 1999;106:1571–8.
- Finger PT, Chin KJ. Antivascular endothelial growth factor bevacizumab for radiation optic neuropathy: secondary to plaque radiotherapy. *Int J Radiation Oncol Biol Phys* 2011 Jan 27 [Epub ahead of print].
- Finger PT, Chin KJ, Yu GP, Palladium-103 for Choroidal Melanoma Study Group. Risk factors for radiation maculopathy after ophthalmic plaque radiation for choroidal melanoma. *Am J Ophthalmol* 2010;149:608–15.
- Finger PT, Chin KJ, Yu GP, Patel NS, Palladium-103 for Choroidal Melanoma Study Group. Risk factors for cataract

- after palladium-103 ophthalmic plaque radiation therapy. *Int J Radiat Oncol Biol Phys* 2011;80:800–6.
27. Levin LA, Gragoudas ES, Lessell S. Endothelial cell loss in irradiated optic nerves. *Ophthalmology* 2000;107:370–4.
28. Char DH, Kroll S, Phillips TL, Quivey JM. Late radiation failures after iodine 125 brachytherapy for uveal melanoma compared with charged-particle (proton or helium ion) therapy. *Ophthalmology* 2002;109:1850–4.
29. Finger PT, Romero JM, Rosen RB, et al. Three-dimensional ultrasonography of choroidal melanoma: localization of radioactive eye plaques. *Arch Ophthalmol* 1998;116:305–12.
30. Garcis JP Jr, Garcia PM, Rosen RB, Finger PT. Optic nerve measurements by 3D ultrasound-based coronal “C-scan” imaging. *Ophthalmic Surg Lasers Imaging* 2005;36:142–6.
31. Tabandeh H, Chaudhry NA, Murray TG, et al. Intraoperative echographic localization of iodine-125 episcleral plaque for brachytherapy of choroidal melanoma. *Am J Ophthalmol* 2000;129:199–204.
32. Jockovich ME, Murray TG, Clifford PD, Moshfeghi AA. Posterior juxtascleral injection of anecortave acetate: magnetic resonance and echographic imaging and localization in rabbit eyes. *Retina* 2007;27:247–52.
33. Pavlin CJ, Japp B, Simpson ER, et al. Ultrasound determination of the relationship of radioactive plaques to the base of choroidal melanomas. *Ophthalmology* 1989;96:538–42.
34. The AJCC Ophthalmic Oncology Task Force. Malignant Melanoma of the Uveal. In: Edge S, Byrd DR, Carducci MA, et al, eds. *AJCC Cancer Staging Manual*. 7th ed. New York: Springer; 2009:547–59.
35. Rivard MJ, Chiu-Tsao ST, Finger PT, et al. Comparison of dose calculation methods for brachytherapy of intraocular tumors. *Med Phys* 2011;38:306–16.
36. Somani S, Sahgal A, Krema H, et al. Stereotactic radiotherapy in the treatment of juxtapapillary choroidal melanoma: 2-year follow-up. *Can J Ophthalmol* 2009;44:61–5.
37. Roberge D, Nordstrom S, Coriveau C, et al. Treatment of medium-sized juxtapapillary melanoma with external Co-60 photon therapy. *Radiother Oncol* 2005;74:71–3.
38. Chess J, Albert DM, Bellows AR, Dallow R. Uveal melanoma: case report of extension through the optic nerve to the surgical margin in the orbital apex. *Br J Ophthalmol* 1984;68:272–5.
39. Finger PT. Anti-VEGF bevacizumab (Avastin) for radiation optic neuropathy. *Am J Ophthalmol* 2007;143:335–8.
40. Finger PT, Chin K. Anti-vascular endothelial growth factor bevacizumab (Avastin) for radiation retinopathy. *Arch Ophthalmol* 2007;125:751–6.
41. Finger PT. Radiation retinopathy is treatable with anti-vascular endothelial growth factor bevacizumab (Avastin). *Int J Radiat Oncol Biol Phys* 2008;70:974–7.
42. Finger PT, Chin KJ. Intravitreal ranibizumab (Lucentis) for radiation maculopathy. *Arch Ophthalmol* 2010;128:249–52.

Footnotes and Financial Disclosures

Originally received: April 27, 2011.

Final revision: July 28, 2011.

Accepted: August 9, 2011.

Available online: November 30, 2011. Manuscript no. 2011-645.

¹ The New York Eye Cancer Center, New York, New York.

² The New York Eye and Ear Infirmary, New York, New York.

³ Beth Israel Comprehensive Cancer Center, New York, New York.

Financial Disclosure(s):

The author(s) have no proprietary or commercial interest in any materials discussed in this article.

This work was supported by The Eye Cancer Foundation, Inc. (<http://eyecancerfoundation.net>). The funding organization had no role in the design or conduct of this research.

Correspondence:

Paul T. Finger, MD, The New York Eye Cancer Center, Suite 5B, 115 East 61st Street, New York, NY 10065. E-mail: pfinger@eyecancer.com.

This article contains online-only material. Tables 1, 2, and 4, and Appendix 1 are available at <http://aaojournal.org>.